Assignment No.4

Title: Use of Predefined Language Models and Analysis of the Impact of Prompt Rephrasing on Model Output

Aim:

To study and analyze how rephrasing a prompt affects the quality, accuracy, and relevance of responses generated by a predefined Large Language Model (LLM).

Theory:

Prompt Engineering is the process of designing and refining input prompts to obtain desired outputs from a language model. Even when the same question is asked, slight changes in wording, structure, or constraints can significantly influence the model's response. This experiment demonstrates how prompt rephrasing improves response quality without changing the underlying model.

Base Question:

Explain Cyber Security.

Prompt Variations and LLM Outputs

Prompt 1: Simple Prompt Explain Cyber Security.

LLM Output:

Cyber Security refers to the practice of protecting computer systems, networks, and data from digital attacks, unauthorized access, and damage. It involves using technologies, processes, and policies to secure sensitive information. Cyber Security is important in preventing data breaches, malware attacks, and identity theft.

Prompt 2: Instruction-Based Prompt

Explain Cyber Security in simple terms suitable for first-year engineering students.

LLM Output:

Cyber Security is the process of protecting computers, mobile devices, and online data from hackers and cyber threats. It ensures that personal and organizational information remains safe and confidential. Examples of cyber security include using strong passwords, installing antivirus software, and securing networks from unauthorized access.

Prompt 3: Role-Based Prompt

You are a network security expert. Explain Cyber Security with real-world examples.

LLM Output:

Cyber Security is the protection of digital systems and networks from cyber threats such as hacking, phishing, and ransomware. For example, banks use encryption to protect customer transactions, and companies use firewalls to block unauthorized access. Email providers use spam filters to prevent phishing attacks. These measures ensure data confidentiality, integrity, and availability.

Prompt 4: Constraint-Based Prompt

Explain Cyber Security in 150 words, including definition, types of threats, and advantages.

LLM Output:

Cyber Security is a field of information technology that focuses on protecting systems, networks, and digital data from cyber threats. It involves the use of security measures such as firewalls, encryption, antivirus software, and multi-factor authentication. Common types of cyber threats include malware, phishing attacks, ransomware, and data breaches. Organizations implement cyber security strategies to protect sensitive information and maintain trust. The advantages of cyber security include prevention of financial loss, protection of personal data, safeguarding of organizational reputation, and ensuring business continuity. In today's digital world, cyber security plays a critical role in maintaining safe online communication and secure data storage.

Prompt 5: Structured Prompt

Explain Cyber Security under the following headings:

1. Definition
2. Common Threats
3. Importance **LLM Output:**

1. Definition:

Cyber Security is the practice of protecting digital systems, networks, and data from unauthorized access and cyber attacks.

2. Common Threats:

Common threats include malware, phishing, ransomware, hacking, and data breaches.

3. Importance:

Cyber Security is important because it protects sensitive information, prevents financial losses, ensures privacy, and maintains trust in digital systems.